JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO AARTHI AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

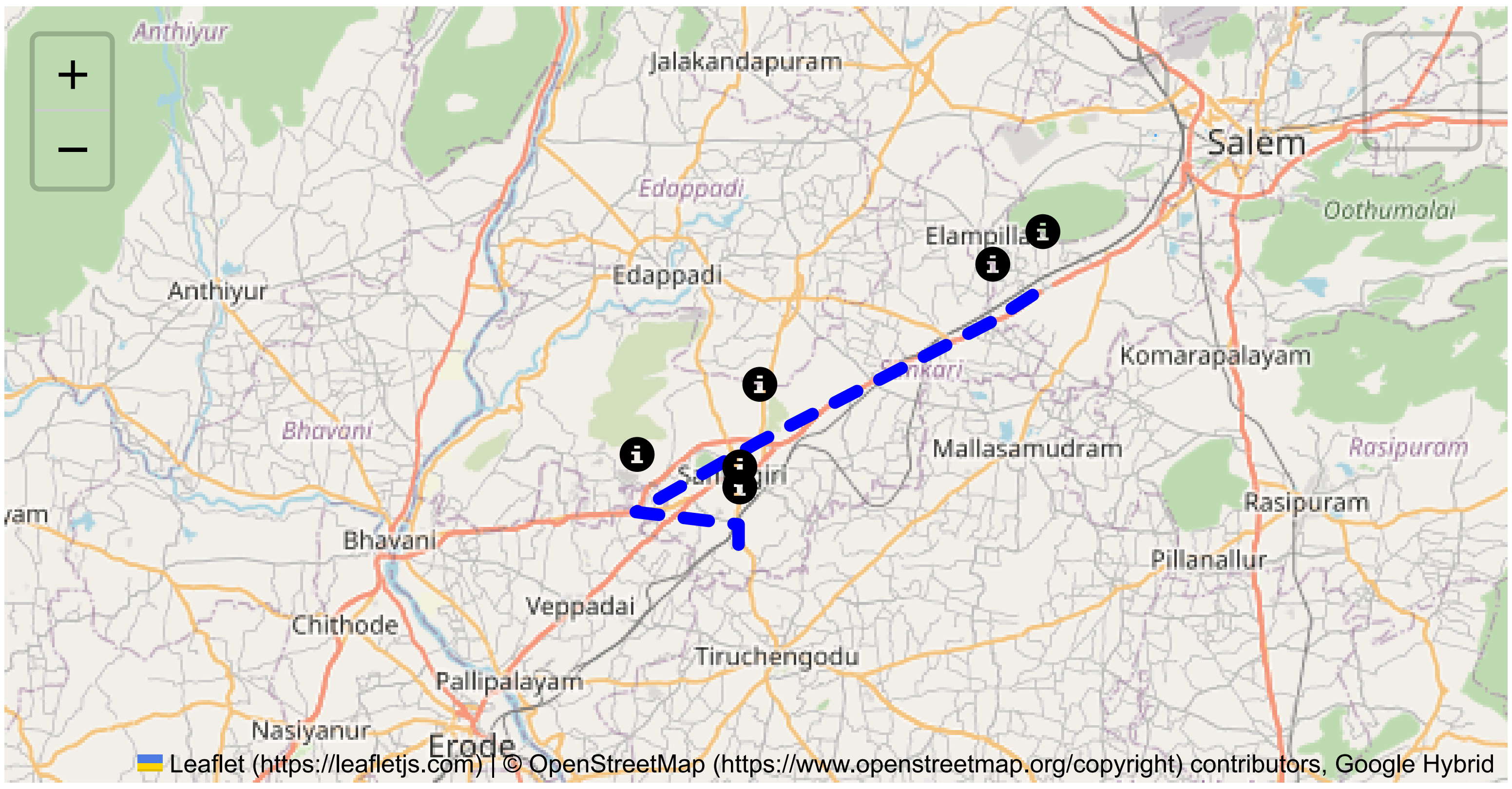
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 37.60 km
Estimated Duration: 0.7 hours
Adjusted Duration (Heavy Vehicle): 0.9 hours
Start: (11.4381, 77.8734)
End: (11.575289, 78.040566)

Welcome to the Journey Risk Management Study

Route Overview:

The route from Sangagiri to Chinnasiragapadi via several key waypoints spans approximately 37.60 kilometers. This route passes through urban and semi-urban areas in Tamil Nadu, predominantly utilizing state highways and regional roads. It is an important corridor connecting several towns, with varied traffic conditions and road quality.

1. Weather Conditions and Hazards:

Tamil Nadu typically has tropical weather. The route experiences high temperatures in the summer (March to June), monsoons (July to September), and mild winters (November to February). The area can experience heavy rain during the monsoon season, which may cause localized flooding and reduced visibility, posing a hazard, particularly in low-lying areas.

2. Traffic Patterns:

Traffic congestion is generally noticed during morning (8:00 AM to 10:00 AM) and evening rush hours (5:30 PM to 8:00 PM). Notable congestion-prone areas include intersections with major highways and urban centers within Sangagiri and Salem districts. Plan travel outside these times to avoid delays.

3. Road Quality and Infrastructure:

The road on this route is a mix of well-maintained highways and some narrow, less maintained local roads. Brisk driving is possible along the Salem - Kochi Highway, but keep in mind possible wear and tear on secondary roads, which might have potholes or uneven surfaces.

4. Alternative Routes:

In case of congestion or emergencies, diverting through major highways like NH44 or NH544 could provide relief. These highways offer alternative routes to bypass localized incidents or heavy traffic.

5. Regulations on Hazardous Materials:

Transport of hazardous materials is subject to regulations by the Motor Vehicles Act, 1988, and any local amendments. Drivers must ensure compliance, including securing permits and displaying appropriate signage. Adherence to prescribed routes is mandatory to minimize risks in densely populated areas.

6. Historical Incidents:

There have been minor incidents of heavy vehicle overturns due to speeding or inadequate road maintenance, particularly on rural stretches. Records of incidents indicate the need for caution in adverse weather and maintaining appropriate speeds.

7. Environmental Considerations:

This route travels through regions near agricultural lands and small industrial areas. Sensitivity must be observed during pesticide applications or harvest times due to agricultural activity, and care with spills, especially near water bodies.

8. Communication Coverage:

While major parts of the route have good mobile coverage, rural stretches might experience occasional dead zones. Ensure to have a reliable GPS system and pre-download offline maps.

9. Emergency Response Times:

Emergency services typically reach urban areas within 30 minutes but may take up to an hour in rural segments, depending on the proximity to the nearest town with emergency facilities.

Overall Risk Assessment Summary:

Transport along this route is moderately risk-prone due to traffic congestion, variable road conditions, and potential for weather-related disruptions. By planning travel during off-peak hours, maintaining communication readiness, and adhering strictly to safety regulations and environmental considerations, risks can be effectively managed.ensko

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44898, 77.87410	30 KM/Hr
5	Turn	Medium	11.45357, 77.85789	30 KM/Hr
6	Turn	Medium	11.45348, 77.85706	30 KM/Hr
7	Turn	Medium	11.45352, 77.85698	30 KM/Hr
8	Turn	Medium	11.46318, 77.84913	30 KM/Hr
9	Turn	High	11.45542, 77.81725	15 KM/Hr
10	Turn	High	11.45631, 77.81668	15 KM/Hr
11	Turn	High	11.49409, 77.88004	15 KM/Hr
12	Turn	Medium	11.49419, 77.88015	30 KM/Hr
13	Turn	Medium	11.49377, 77.88573	30 KM/Hr
14	Turn	High	11.49363, 77.88590	15 KM/Hr
15	Turn	High	11.49235, 77.89561	15 KM/Hr
16	Blind Spot	Blind Spot	11.49225, 77.89560	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44898, 77.87410

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45357, 77.85789

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45348, 77.85706

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45352, 77.85698

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

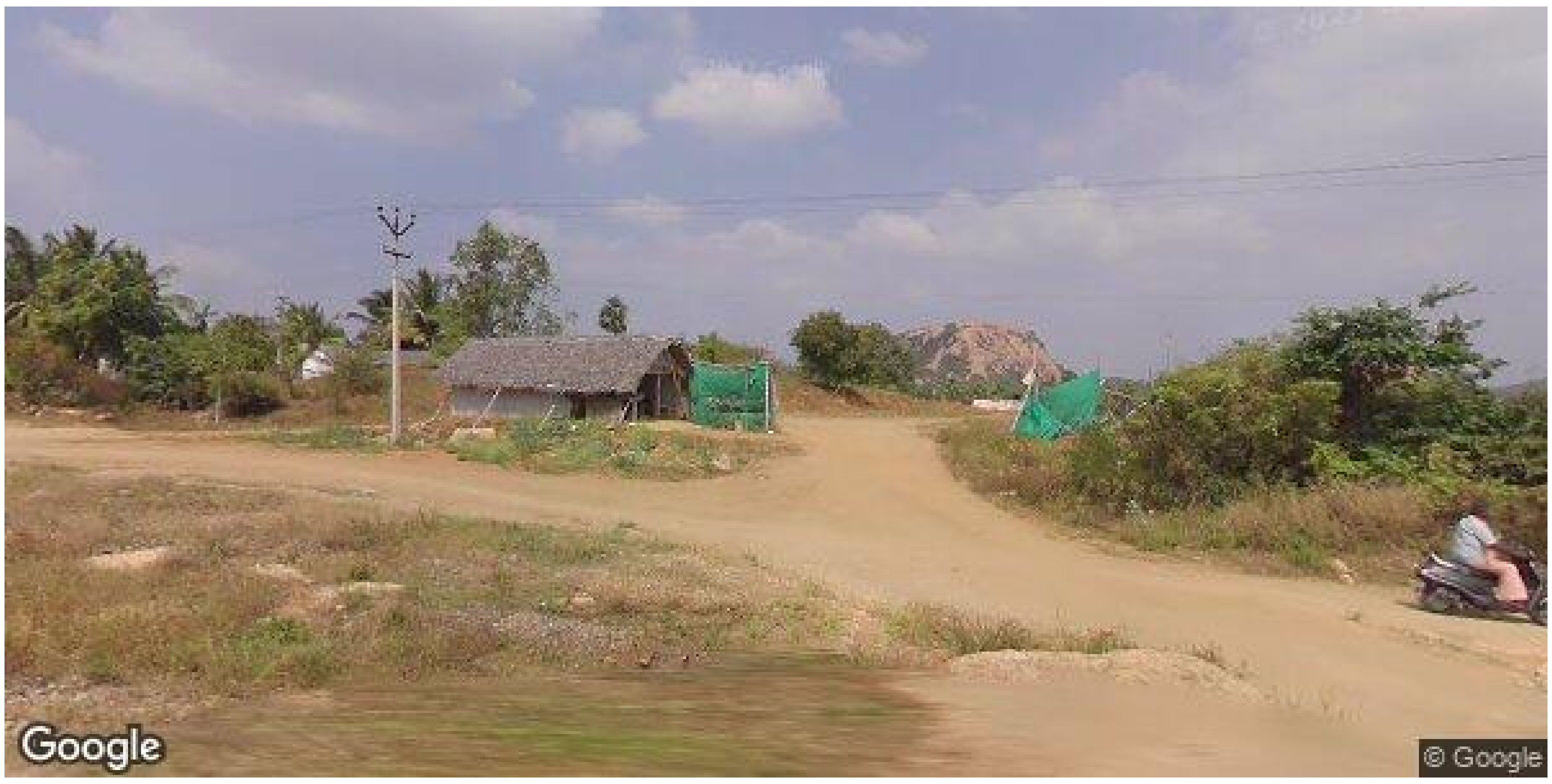
Coordinates: 11.45631, 77.81668

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49409, 77.88004



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49419, 77.88015

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49377, 77.88573

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49363, 77.88590

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49235, 77.89561

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49225, 77.89560